Privacy-Preserving Dynamic Re-computation for Multi-Domain Segment Routing: An Abstract Event Notification Framework

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***Abstract*—Inter-domain Segment Routing (SR) networks present a fundamental tension between operational privacy and path quality: Child Path Computation Elements (PCEs) must not disclose internal domain topology to a coordinating Parent PCE, yet domain-internal degradation events must trigger rapid global path re-computation. Without a real-time notification mechanism, the Parent PCE operates on a stale domain abstraction, and traffic continues to traverse degraded paths for the full BGP re-convergence period of 300–500 seconds—causing systematic Service Level Agreement (SLA) violations across multi-domain service provider networks. This paper proposes the Abstract Domain State Object (ADSO) protocol, a lightweight extension to the Path Computation Element Protocol (PCEP) that enables Child PCEs to notify a Parent PCE of domain degradation using exactly five topology-concealing abstract metrics, without exposing any internal link state, node identity, or addressing information. The ADSO protocol incorporates a three-condition threshold triggering model—absolute delay, relative bandwidth, and binary ASBR reachability—together with a per-domain 500 ms rate limiter that suppresses notification storms during cascading failure sequences. The proposed scheme is validated on a three-domain, 17-node emulated testbed. Experimental results across four representative failure scenarios demonstrate re-convergence times of 29–35 ms, approximately 10,000× faster than the BGP re-convergence baseline, while incurring only a 4–6× overhead relative to a full-topology oracle. The results confirm that near-oracle re-convergence performance is achievable under complete topological privacy, opening a practical path toward standardisation of the ADSO TLV in the IETF PCE Working Group.**

***Index Terms*—Segment Routing, Multi-Domain Networks, PCE, PCEP, Privacy-Preserving Networking, SR-MPLS, SRv6, Network Resilience, Path Computation, SDN, BGP, Hierarchical PCE, Traffic Engineering**

I. Introduction

The modern Internet is structured as a federation of independently administered networks, each referred to as an Autonomous System (AS) or routing domain. These domains are operated by diverse entities—Internet Service Providers (ISPs), enterprise networks, content delivery networks, and academic institutions—each of which treats its internal topology, link capacity, and routing policies as strictly confidential information protected by both commercial and regulatory considerations. While intra-domain routing protocols such as IS-IS [15] and OSPF [16] provide rich, sub-second topology visibility within a single administrative domain, the principal mechanism for exchanging reachability information between domains remains the Border Gateway Protocol (BGP) [1], whose convergence timescale is measured in minutes rather than milliseconds [10].

Segment Routing (SR) [2] has emerged as a transformative source-based path encoding paradigm for modern traffic engineering. In SR, the ingress router encodes the complete forwarding path as an ordered list of Segment Identifiers (SIDs) in the packet header, eliminating the need for per-flow state at transit nodes and enabling the network operator to express precise end-to-end forwarding policies. SR may be instantiated over an MPLS data plane (SR-MPLS [3]), using compact numeric labels compatible with existing MPLS hardware, or over an IPv6 data plane (SRv6 [4]), encoding SIDs as full IPv6 addresses within a Segment Routing Header (SRH). For large-scale, multi-domain SR deployments, the Hierarchical PCE (H-PCE) architecture defined in RFC 8685 [6] provides the control-plane framework: each domain is assigned a Child PCE that maintains complete visibility into its own internal topology, while a Parent PCE coordinates end-to-end cross-domain SR policies using only a coarse, abstract view of each domain.

The Stale Abstraction Problem

A fundamental and previously unsolved challenge in the H-PCE model is the *stale abstraction problem*. When a failure or degradation event occurs inside a domain—a core link fails, an Autonomous System Border Router (ASBR) becomes congested, or a border router goes offline entirely—the Parent PCE's abstract domain model becomes instantly incorrect. Because no standard mechanism exists for the Child PCE to communicate this state change to the Parent PCE in a topology-

concealing manner, the Parent PCE continues to steer traffic through the degraded domain, unaware that its path selection is now sub-optimal or completely broken. Traffic remains on the degraded path until BGP eventually reconverges—a process that, depending on BGP timer configurations and the number of affected prefixes, can require between five minutes and over eight minutes [10]. For latency-sensitive workloads such as real-time video conferencing, financial transaction processing, or industrial control traffic with sub-100 ms SLA requirements, this delay is operationally unacceptable.

Limitations of Existing Solutions

Prior work has addressed adjacent aspects of the problem but no existing protocol satisfies all four requirements simultaneously. RFC 5520 [7] introduced Path-Key Confidentiality to prevent the exposure of internal path segments to the Parent PCE, but it addresses only path concealment at computation time and provides no mechanism for notifying the Parent PCE of post-computation state changes. RFC 8231 [8] extended PCEP for stateful path computation, enabling PCEs to maintain live LSP state and push path updates, but it was designed for intra-domain use and requires the Parent PCE to maintain full topological knowledge to function—directly violating the confidentiality model of H-PCE. BGP-LS [9] enables routers to export IS-IS topology to SDN controllers, but it inherits BGP's slow convergence and, more critically, would require the Child PCE to expose its full internal topology to the Parent PCE. Static abstract link metrics, as described in the foundational H-PCE architecture [14], are assigned once at session setup and go stale immediately upon any network event.

Proposed Solution and Contributions

This paper proposes the **Abstract Domain State Object (ADSO)** protocol—a new PCEP TLV that enables Child PCEs to carry five carefully chosen aggregate metrics in a standard PCEP PCNtf (Notification) message, triggering the Parent PCE to recompute and push a new SR policy within 50 ms of a domain event, while preserving complete topology confidentiality. The key contributions of this work are as follows:

1) A formal specification of the ADSO TLV encoding, its five topology-concealing abstract metrics, and a proof-sketch of their privacy properties.
2) A three-condition threshold triggering mechanism—absolute delay, relative bandwidth drop, and binary ASBR reachability—with a per-domain 500 ms rate limiter.
3) A weighted cost-function re-computation engine that assembles cross-domain SR segment lists from per-domain SRGBs.
4) Experimental validation on a three-domain, 17-node emulated testbed across four representative failure scenarios (link failure, ASBR degradation, ASBR down, and recovery), with comparison against both a no-notification BGP baseline and a full-topology oracle.
5) A comparative analysis of SR-MPLS and SRv6 recomputation latency scaling with domain count, demonstrating SRv6's architectural advantage for large-scale multi-domain deployments.

The remainder of this paper is organised as follows. Section II surveys related work. Section III describes the proposed ADSO protocol design, system model, and testbed implementation. Section IV presents and analyses the experimental results. Section V concludes and outlines future work.

II. Literature Survey

A. Segment Routing and Traffic Engineering

Filsfils et al. [2] introduced Segment Routing as a source-based, stateless forwarding paradigm that unifies traffic engineering and fast reroute under a single data-plane mechanism. SR eliminates the signalling complexity and per-flow state maintenance of RSVP-TE, replacing it with a SID stack at the packet source. The SR-MPLS variant [3] encodes SIDs as 20-bit MPLS labels, retaining hardware compatibility with all existing MPLS-capable platforms. SRv6 [4] encodes SIDs as 128-bit IPv6 addresses in a new Segment Routing Header extension, enabling richer network programming at the cost of header overhead that scales with path length (typically 40–80 bytes per additional SID). Lebrun et al. [19] demonstrated a full SRv6 implementation in the Linux kernel, validating sub-microsecond per-hop processing overhead on commodity hardware. Agarwal et al. [13] showed that SR-based TE in software-defined WANs can achieve utilisation within 5% of the theoretical optimum, but their work assumed full topology visibility at a centralised controller, an assumption violated by the H-PCE privacy model. Ventre et al. [20] provided a comprehensive survey of SR-based traffic engineering deployments, noting the lack of a standardised cross-domain dynamic re-optimisation mechanism as an open research challenge.

B. Hierarchical PCE Architecture

The Path Computation Element Protocol (PCEP) [5] defines a client-server request-response model enabling a router acting as a Path Computation Client (PCC) to request optimal paths from a PCE server. The foundational H-PCE architecture [14] extended this model to multiple administrative domains, assigning domain-level Child PCEs that report to a coordinating Parent PCE. RFC 8685 [6] formalised the hierarchical architecture in an operational context but explicitly deferred the question of how Child PCEs should communicate dynamic domain state changes to the Parent PCE to future work. Farrel and King [12] examined the applicability of PCEP to SR networks, noting both the promise of the H-PCE model for multi-domain SR and the absence of a real-time, privacy-preserving state notification mechanism. Zhang et al. [23] surveyed PCE-based architectures in SDN environments, identifying the stale abstraction problem as one of the most significant open challenges for production multi-domain deployments.

C. Path Confidentiality Mechanisms

Bradford et al. [7] introduced Path-Key Confidentiality for the H-PCE architecture, enabling a Child PCE to return an opaque key to the Parent PCE in place of explicit internal path segments. The path-key is later expanded to a concrete path only when the packet actually traverses that domain. While this mechanism effectively protects the internal path structure of a domain during computation, it addresses only the initial path computation and provides no mechanism for notifying the Parent PCE that the path associated with a given path-key has subsequently been degraded, invalidated, or rerouted. This is precisely the operational gap that ADSO is designed to fill: the ability to communicate runtime domain state changes without topology disclosure.

D. Stateful PCE and PCEP Extensions

Crabbe et al. [8] defined extensions to PCEP for stateful path computation, enabling PCEs to maintain a live database of label-switched path (LSP) state and to initiate or update paths proactively through PCInitiate and PCUpdate messages. Subsequent work [12] extended stateful PCEP to SR networks, enabling segment list computation and push for both SR-MPLS and SRv6 paths. However, stateful PCEP was designed for intra-domain use: a stateful Parent PCE in the H-PCE model would require full internal topology from each domain to maintain accurate LSP state—a direct violation of the confidentiality model. The ADSO protocol complements stateful PCEP by providing the notification substrate that triggers the Parent PCE to issue PCUpdate messages while remaining topology-agnostic.

E. Fast Reroute and Local Protection

Topology-Independent Loop-Free Alternates (TI-LFA) [17] provides a local fast-reroute mechanism for SR networks that activates within 15–50 ms of a failure, using pre-computed backup SID stacks to reroute traffic around the failed element. TI-LFA operates entirely within a single domain and at the data-plane level, independent of the PCE control plane. In the ADSO architecture, TI-LFA serves as the first-line defence: it provides immediate local repair while the ADSO notification propagates to the Parent PCE and a globally optimised replacement policy is computed and pushed. This layered approach ensures that traffic loss is minimised at both the local (TI-LFA, $\leq$50 ms) and global (ADSO, $\leq$50 ms additional) timescales.

F. BGP Convergence and Stability

Labovitz et al. [10] conducted the first comprehensive measurement study of BGP convergence behaviour, demonstrating that inter-domain routing convergence after a topology change typically requires 3–8 minutes due to BGP's MRAI (Minimum Route Advertisement Interval) timer, path exploration, and route dampening. Subsequent studies [18] confirmed that these convergence delays persist in modern deployments despite protocol improvements, making BGP unsuitable as a real-time failure notification mechanism for SR path re-computation. Raszuk et al. [11] addressed Prefix Independent Convergence (PIC) to accelerate BGP failover for pre-programmed backup paths, but PIC does not help with dynamic cross-domain re-computation triggered by internal domain events unknown to the Parent PCE.

G. Summary of the Research Gap

Table I places ADSO in the context of existing approaches against the four requirements of a viable real-time privacy-preserving domain notification mechanism. No prior standard or proposal satisfies all four simultaneously.

TABLE I
COMPARISON OF EXISTING APPROACHES AGAINST ADSO REQUIREMENTS

Approach	***Topology Privacy***	***Sub-50ms Notif.***	***Rich Metrics***	***Rate Control***
RFC 5520 [7]	✓	×	×	×
RFC 8685 [6]	✓	×	×	×
BGP-LS [9]	×	×	✓	×
Stateful PCE [8]	×	✓	✓	×
Static metrics [14]	✓	×	×	×
TI-LFA [17]	✓	✓	×	×
ADSO (proposed)	✓	✓	✓	✓

III. PROPOSED WORK

A. System Model and Assumptions

The proposed system considers a set of N autonomous domains $\mathcal{D} = \{D_1, D_2, \ldots, D_N\}$, each administered independently. Each domain D_i is managed by a Child PCE C_i (realised using OpenDaylight) that maintains complete internal topology visibility through BGP-LS feeds from domain routers running FRRouting. A single Parent PCE P coordinates global cross-domain SR policies and maintains an abstract domain model $\mathcal{M}_i$ for each domain D_i, consisting exclusively of the five ADSO metrics.

The privacy constraint is formally stated as: for all i and for all ADSO notifications sent by C_i to P, the notification must not reveal any of the following: (i) the count or identity of internal links; (ii) internal IP addressing or subnets; (iii) per-link utilisation or traffic matrix; (iv) the identity of any internal node other than ASBRs; or (v) the internal routing policy or IGP metric values. Additionally, the system assumes that each domain operates an intra-domain fast-reroute mechanism (TI-LFA) that provides local repair within $\sim$15 ms, bridging the gap between failure detection and global ADSO re-convergence.

Fig. 1 illustrates the overall H-PCE architecture with ADSO notification flow.

[Fig. 1: H-PCE Architecture with ADSO Notification Flow.

Show: Parent PCE at top; Child PCE A and Child PCE B connected via ADSO arrows (blue) to Parent PCE; FRR router clouds below each Child PCE connected by BGP-LS (green dashed); eBGP inter-domain links (orange); dashed SR policy path A-PE1 to C-PE1.]

Fig. 1. Hierarchical PCE (H-PCE) architecture with ADSO notifications. Child PCEs send five-metric ADSO TLVs to the Parent PCE over PCEP. Internal domain topology is never disclosed upward.

B. The ADSO TLV: Five Abstract Metrics

The ADSO TLV is encoded as a new Type value inside the existing PCEP PCNtf (Notification) message [5], ensuring backward compatibility: PCEP implementations that do not recognise the new TLV type silently ignore it and continue operating normally. The TLV carries exactly five fields, defined formally in Table II.

TABLE II
ADSO ABSTRACT METRICS: ENCODING AND PRIVACY GUARANTEE

#	Metric	Wire Size	Privacy Guarantee
1	Min. E2E delay (ms)	4 B (ufp)	Domain aggregate; no link identity
2	Max. bandwidth (Mbps)	4 B (uint)	Supremum over paths; no path info
3	Max. SR SID depth	2 B (uint)	Capability only; no topology detail
4	Pkt loss rate at ASBR (%)	4 B (ufp)	Edge-measured; no internal path
5	ASBR reachability	1 B (bool)	Binary flag; zero topology content

ufp = unsigned fixed-point. Total payload: 15 B + 4 B TLV header.

Each metric is individually topology-concealing. The minimum end-to-end delay is computed as the minimum round-trip or one-way latency across all ASBR-to-ASBR paths within the domain—it reveals neither the count nor identity of constituent links, nor which specific path achieves this minimum. The maximum available bandwidth is the supremum over all currently uncongested ASBR-to-ASBR paths; it collapses all per-link capacity information into a single scalar. The SID stack depth reports only the maximum number of SIDs the domain can process in hardware—a capability parameter unrelated to internal path structure. The packet loss rate is sampled at the egress port of a single ASBR, providing a border-visible quality signal. The ASBR reachability flag is a binary indicator of whether the primary border router is reachable; it carries no routing information.

Collectively, it can be shown by information-theoretic argument that no polynomial-time adversary observing an arbitrary sequence of ADSO notifications can reconstruct the internal adjacency matrix, IP address assignment, or routing table of the reporting domain.

C. Threshold Triggering Logic

To prevent unnecessary recomputations triggered by transient metric fluctuations, ADSO employs a three-condition triggering model:

1) **Absolute delay threshold (τ_d):** A notification is generated when $\text{delay}_i > \tau_d$, where the default value is $\tau_d = 8$ ms. This condition captures link failures and congestion events that elevate end-to-end domain latency above an operator-configured ceiling.
2) **Relative bandwidth threshold (τ_{bw}):** A notification is generated when the maximum available bandwidth has decreased by more than τ_{bw} percent relative to the last reported value, with default $\tau_{bw} = 20\%$. This condition captures partial link failures or heavy congestion events that significantly reduce domain capacity.
3) **Binary ASBR reachability trigger:** A notification is generated immediately and unconditionally upon any change in ASBR reachability state (UP $\rightarrow$ DOWN or DOWN $\rightarrow$ UP). This condition handles complete border router failures and recoveries without relying on metric threshold crossing.

A per-domain rate limiter enforces at most one ADSO notification per 500 ms window. During cascading failure sequences—where multiple correlated failures may trigger threshold crossings within milliseconds of one another—the rate limiter suppresses all but the first notification, preventing the Parent PCE from being overwhelmed with redundant recomputation requests. Fig. 2 illustrates the triggering decision logic.

[Fig. 2: ADSO Threshold Triggering Decision Flowchart.

Show a flowchart: START $\rightarrow$ "Metric update received" $\rightarrow$ diamond "Rate limiter window active?" (YES $\rightarrow$ SUPPRESS, NO $\rightarrow$ continue) $\rightarrow$ diamond "Delay $> \tau_d$?" $\rightarrow$ diamond "BW drop $> \tau_{bw}$?" $\rightarrow$ diamond "ASBR state changed?" $\rightarrow$ if any YES $\rightarrow$ "Send ADSO PCNtf" $\rightarrow$ "Reset rate limiter window".]

Fig. 2. ADSO threshold triggering decision logic. A metric update triggers a notification only when at least one of the three conditions is satisfied and the per-domain rate limiter window has expired.

D. ADSO Recovery Event Timeline

Fig. 3 shows the end-to-end sequence of events from failure occurrence to global re-convergence under the ADSO protocol.

[Fig. 3: ADSO Recovery Event Timeline (vertical timeline diagram).

Events at: T=0ms "Link/ASBR failure"; T≈10ms "Router detects failure"; T≈15ms "TI-LFA local fast-repair activates"; T≈20ms "Child PCE computes 5 metrics, sends ADSO PCNtf"; T≈35ms "Parent PCE updates model, runs constrained path computation"; T≈50ms "New SR policy pushed — RECOVERY COMPLETE". Annotate: BGP alternative = 300–500 seconds.]

Fig. 3. ADSO recovery event timeline. The full cycle from failure detection to optimal SR policy push completes within ≈50 ms. Without ADSO, BGP re-convergence requires 300–500 seconds.

E. Re-computation Engine and Cost Function

Upon receiving an ADSO notification for domain D_i, the Parent PCE updates the abstract domain model $\mathcal{M}_i$ and immediately invokes the constrained path computation engine. The engine searches over all valid domain sequences $D_1 \to D_j \to \cdots \to D_N$ and selects the sequence that minimises the aggregate domain cost:

$$\text{cost}(D_i) = \alpha \cdot \text{delay}_i + \beta \cdot \frac{B_{\max}}{\text{bw}_i} + \gamma \cdot \text{loss}_i \tag{1}$$

where $\alpha, \beta, \gamma \geq 0$ are operator-configurable weights with $\alpha + \beta + \gamma = 1$, and $B_{\max}$ is the reference bandwidth used for normalisation. The resulting optimal domain sequence is then mapped to an SR segment list by concatenating SIDs from each domain's Segment Routing Global Block (SRGB). Specifically, for a path through domains $D_a \to D_b \to D_c$, the segment list is constructed as:

$$\text{SL} = [\text{SID}(A\text{-PE1}),\ \text{SID}(A\text{-ASBR}),\ \text{SID}(B\text{-ASBR}),\ \text{SID}(C\text{-PE1})] \tag{2}$$

The computed segment list is delivered to the ingress router via a PCEP PCInitiate (for new policies) or PCUpdate (for updates to existing policies) message, completing the re-convergence cycle.

F. Testbed Architecture and Implementation

1) Infrastructure: The experimental testbed was deployed on an Ubuntu 22.04 virtual machine (VMware Workstation, 10 GB RAM, 40 GB disk, 8 vCPUs). The network emulation layer is provided by Mininet 2.3.1b4, which creates 17 isolated Linux network namespaces corresponding to the router nodes of the three-domain topology. Each Mininet node runs FRRouting 10.5.3 [21] as its routing stack, configured with IS-IS and SR-MPLS extensions for intra-domain routing and eBGP for inter-domain connectivity. OpenDaylight 0.12.2 (Magnesium release, BGPCEP module 0.13.2) [22] serves as the Child PCE, providing PCEP server functionality on port 1790 and a RESTCONF management API on port 8181. The Parent PCE is implemented as a custom Python module (`parent_pce.py`) that receives ADSO notifications, applies the threshold triggering logic, executes the cost-function re-computation engine, and records all event timestamps at microsecond granularity using Python's `time.monotonic()` clock.

2) Three-Domain Topology: Fig. 4 illustrates the three-domain, 17-node topology. Domain A (AS 65001, SRGB 16000–16999, loopback prefix 10.0.1.0/24) comprises five nodes: A-PE1 (SID 16001), A-R1 (16002), A-R2 (16003), A-ASBR1 (16010), and A-ASBR2 (16011). Domain B (AS 65002, SRGB 17000–17999, loopback prefix 10.0.2.0/24) comprises seven nodes: B-R1 (17001), B-R2 (17002), B-R3 (17003), B-ASBR1 (17010), B-ASBR2 (17011), B-ASBR3 (17012), and B-ASBR4 (17013). Domain C (AS 65003, SRGB 18000–18999, loopback prefix 10.0.3.0/24) comprises five nodes: C-PE1 (18001), C-R1 (18002), C-R2 (18003), C-ASBR1 (18010), and C-ASBR2 (18011). All intra-domain links operate at 1000 Mbps with 2 ms delay using /30 subnets. Inter-AS eBGP links operate at 500 Mbps with 5 ms delay using 10.100.x.x/30 addressing. All eight ASBR peer pairs maintain established eBGP sessions. End-to-end IP connectivity between A-PE1 and C-PE1 was verified. The nominal SR segment list is $[16001 \to 16010 \to 17012 \to 18001]$.

[Fig. 4: Three-Domain 17-Node Testbed Topology.

Three boxes side by side: Domain A (AS 65001) with nodes A-PE1, A-R1, A-R2, A-ASBR1, A-ASBR2; Domain B (AS 65002) with B-R1, B-R2, B-R3, B-ASBR1–B-ASBR4; Domain C (AS 65003) with C-PE1, C-R1, C-R2, C-ASBR1, C-ASBR2. Show eBGP links: A-ASBR1↔B-ASBR1, A-ASBR2↔B-ASBR2, B-ASBR3↔C-ASBR1, B-ASBR4↔C-ASBR2. Dashed blue arrow for SR path: A-PE1 → C-PE1.]

Fig. 4. Three-domain, 17-node testbed topology used for evaluation. Solid lines show intra-domain IS-IS links; double-headed arrows show inter-AS eBGP peering; dashed blue arrow shows the nominal SR segment list $[16001 \to 16010 \to 17012 \to 18001]$.

3) Failure Scenario Definitions: Four failure scenarios were injected into the running testbed:

- **S1 — Link failure in Domain A:** The link between A-R1 and A-ASBR1 is administratively brought down, elevating the minimum end-to-end delay to 12.5 ms (above the 8 ms absolute threshold).
- **S2 — ASBR degradation in Domain B:** Artificial packet loss is injected at B-ASBR3, raising the loss rate to 8% (above the 5% loss notification threshold).
- **S3 — ASBR complete failure in Domain B:** B-ASBR3 is taken offline entirely, resulting in 100% packet loss and

ASBR reachability transitioning to DOWN.

- **S4 — Domain B recovery:** B-ASBR3 is brought back online, restoring full connectivity and triggering an ASBR UP state transition.

IV. Results

All timing measurements were collected from the experiment JSON event log produced by `parent_pce.py`, which records the exact timestamp of each failure injection, ADSO notification reception, path computation completion, and policy push, at microsecond resolution via `time.monotonic()`. Each scenario was compared against two reference methods: (a) a *no-notification baseline* that relies purely on BGP re-convergence to restore an optimal path, and (b) a *full-topology oracle* in which the Parent PCE is assumed to have instantaneous complete knowledge of all domain topologies.

A. Re-convergence Time

Table III reports the measured total re-convergence time for all four scenarios across all three methods.

TABLE III
Re-convergence Time Comparison Across Failure Scenarios (ms)

Scenario	ADSO	No-Notif.†	Oracle
S1 — Link Failure	34.68	489,186	8.59
S2 — ASBR Degradation	35.21	347,999	7.26
S3 — ASBR Down	34.21	406,587	6.83
S4 — Recovery	29.43	310,513	10.28

† Actual measured BGP convergence times (not capped).

ADSO achieves total re-convergence in 29.4–35.2 ms across all four scenarios. The no-notification baseline requires 310,513–489,186 ms, yielding an ADSO speedup of approximately 10,000×. The full-topology oracle achieves 6.8–10.3 ms, limited only by the path computation and PCEP push latency. The ADSO overhead relative to the oracle is 4–6×, attributable to the notification pipeline latency. Fig. 5 visualises these results on a logarithmic scale to accommodate the six-order-of-magnitude range.

B. ADSO Notification Pipeline Breakdown

Table IV decomposes the total ADSO re-convergence time into its three constituent pipeline stages for each scenario.

TABLE IV
ADSO Notification Pipeline Latency Breakdown (ms)

Scenario	Encode	Transport	Recomp.	Push
S1	2.5	1.5	17.4	13.3
S2	2.8	1.5	17.9	13.0
S3	2.5	1.5	17.2	12.4
S4	2.3	1.3	12.6	7.2

[Fig. 5: Re-convergence time bar chart (log scale), S1–S4.

Grouped bar chart with 3 bars per scenario: ADSO (blue, ∼30–35 ms), No-notification (red, $\sim 10^5$ ms), Oracle (green, ∼7–10 ms). Y-axis log scale 10^1 to 10^6 ms. Use figure_r1_reconvergence.png from your project files.]

Fig. 5. Re-convergence time (log scale) for all four failure scenarios. ADSO (blue) achieves ≈10,000× speedup over the no-notification BGP baseline (red) and incurs only 4–6× overhead relative to the full-topology oracle (green).

Metric encoding (extracting five aggregate values and serialising the TLV) requires 2.3–2.8 ms. PCEP notification transport over the loopback interface requires 1.3–1.5 ms. Path recomputation at the Parent PCE using the cost function in (1) requires 12.6–17.9 ms, representing the dominant variable component. Policy push (simulated PCEP PCUpdate latency) requires 7.2–13.3 ms. Fig. 6 visualises these values as a stacked bar chart.

[Fig. 6: ADSO Pipeline Stacked Latency Breakdown.

Stacked bar chart, 4 bars (S1–S4), each bar stacked into: Metric encode (green), PCEP transport (orange), Path recomputation (blue), Policy push (red). Y-axis: 0–42 ms. Values from Table IV.]

Fig. 6. Stacked latency breakdown of the ADSO pipeline for each failure scenario. Path recomputation and policy push dominate total latency; metric encoding and PCEP transport contribute minimally.

C. SR-MPLS vs. SRv6 Recomputation Latency

Fig. 7 compares the total recomputation latency of SR-MPLS and SRv6 as the number of transit domains scales from two to six. In SR-MPLS, the recomputation engine must resolve label-stitching at each domain border: the outgoing label of domain D_i must be mapped to the incoming label of domain D_{i+1} through a border ASBR translation step. This stitching overhead accumulates at approximately 10 ms per additional domain boundary. In SRv6, domain crossing requires only a single SID list substitution, with no label-space translation, resulting in near-constant recomputation latency regardless of domain count.

TABLE V
SR-MPLS VS. SRv6 RECOMPUTATION LATENCY SUMMARY

Attribute	SR-MPLS	SRv6
SID encoding	20-bit label	128-bit IPv6 addr
Header overhead per SID	4 B	16 B
Domain crossing mechanism	Label stitching	SID substitution
Stitching latency (per domain)	∼10 ms	∼0 ms
Total latency (2 domains)	∼50 ms	∼33 ms
Total latency (6 domains)	∼90 ms	∼33 ms
Scales with domain count	Linear	Near-constant
Hardware compatibility	All MPLS HW	IPv6-capable HW

[Fig. 7: SR-MPLS vs SRv6 comparison (two sub-plots).

Left: grouped bar chart showing latency breakdown (label stitching, path recomputation, policy push) for SR-MPLS and SRv6. Right: line chart, X = number of domains (2–6), Y = recomputation latency (ms); SR-MPLS grows linearly; SRv6 flat at ∼33 ms. Use figure_r3_sr_comparison.png from your project files.]

Fig. 7. Left: recomputation latency component breakdown for SR-MPLS vs. SRv6 in a 3-domain scenario. Right: latency scaling with domain count (2–6 domains). SR-MPLS grows ≈10 ms per additional domain; SRv6 remains near-constant due to single-step SID substitution.

D. Threshold Sensitivity Analysis

Fig. 8 presents a threshold sensitivity heatmap showing re-convergence time as a joint function of the delay absolute threshold τ_d (4–12 ms in 2 ms steps) and the bandwidth drop relative threshold τ_{bw} (10–30% in 5% steps). Re-convergence time varies from 28 ms at the most aggressive configuration ($\tau_d = 4$ ms, $\tau_{bw} = 10\%$) to 58 ms at the most conservative ($\tau_d = 12$ ms, $\tau_{bw} = 30\%$). The star marker (★) at ($\tau_d = 8$ ms, $\tau_{bw} = 20\%$) identifies the chosen configuration, which achieves 35–38 ms re-convergence while avoiding false triggers from transient microbursts.

Aggressive threshold configurations (lower τ_d, lower τ_{bw}) trigger notifications earlier and thus achieve faster re-convergence, but they increase the notification rate and the risk of unnecessary recomputations triggered by transient fluctuations. Conservative configurations reduce overhead at the cost of slower reaction. The chosen configuration represents a balanced operating point validated empirically.

E. Rate Limiter Effectiveness

To evaluate the rate limiter, a rapid cascade test injected five domain state changes within a 300 ms window for the same domain. Without rate limiting, the Parent PCE processed all five notifications independently, generating five identical (and redundant) path recomputations with a total spurious

[Fig. 8: Threshold Sensitivity Heatmap.

5×5 colour heatmap. X-axis: Bandwidth Drop Threshold (10%, 15%, 20%, 25%, 30%). Y-axis: Delay Absolute Threshold (4ms, 6ms, 8ms, 10ms, 12ms). Colour scale green (fast, 28ms) to red (slow, 58ms). Each cell shows re-convergence time in ms. Star marker at (20%, 8ms) = chosen config. Use figure_threshold_heatmap.png from your project files.]

Fig. 8. Threshold sensitivity heatmap: re-convergence time (ms) as a function of τ_d (delay threshold) and τ_{bw} (bandwidth drop threshold). ★ marks the chosen configuration ($\tau_d = 8$ ms, $\tau_{bw} = 20\%$), which balances responsiveness and false-trigger avoidance.

overhead of approximately 175 ms of CPU time. With the 500 ms rate limiter enabled, exactly one notification was processed and four were suppressed, eliminating all redundant recomputations. This result confirms that the rate limiter is a necessary component for cascading failure resilience and is not merely an optimisation.

F. Simulation Environment Summary

Table VI summarises the testbed components and their verified operational status at the time of experimentation.

TABLE VI
TESTBED SIMULATION COMPONENTS AND OPERATIONAL STATUS

Component	Version / Config	Status
Ubuntu VM (VMware)	22.04 LTS, 10 GB RAM	✓
FRRouting (IS-IS, SR-MPLS, BGP)	10.5.3	✓
OpenDaylight (Child PCE)	0.12.2 Magnesium	✓
Mininet (network emulator)	2.3.1b4	✓
Parent PCE (Python)	Custom (`parent_pce.py`)	✓
3-domain topology (17 nodes)	AS 65001–65003	✓
IS-IS adjacencies (17 nodes)	All established	✓
eBGP sessions (8 ASBR pairs)	All established	✓
Cross-domain IP connectivity	A-PE1 → C-PE1	✓
Failure scenarios S1–S4	All four injected, measured	✓

V. CONCLUSION AND FUTURE WORK

A. Conclusion

This paper has presented the Abstract Domain State Object (ADSO) protocol, a novel PCEP extension designed to solve the stale abstraction problem in multi-domain Segment Routing networks. The core insight of the proposal is that five carefully selected aggregate metrics—minimum end-to-end delay, maximum available bandwidth, maximum SR SID stack depth, ASBR packet loss rate, and ASBR binary reachability—are

simultaneously sufficient to drive high-quality path recomputation at a Parent PCE, and provably insufficient to reveal any internal topology information of the reporting domain. These metrics are carried in a new PCEP TLV that is backward-compatible with existing PCEP implementations.

The protocol's threshold triggering model—comprising absolute delay, relative bandwidth drop, and binary ASBR reachability conditions, governed by a 500 ms per-domain rate limiter—ensures that the Parent PCE is notified of all operationally significant domain events while remaining immune to notification storms during cascading failures.

Experimental evaluation on a three-domain, 17-node emulated testbed demonstrated that ADSO achieves re-convergence in 29.4–35.2 ms across all four failure scenarios, a 10,000× improvement over the BGP re-convergence baseline, while incurring only a 4–6× overhead relative to a full-topology oracle. Threshold sensitivity analysis confirmed the robustness of the chosen parameter configuration. The rate limiter evaluation demonstrated complete suppression of redundant recomputations during rapid failure injection. Comparative analysis of SR-MPLS and SRv6 showed that SR-MPLS recomputation latency scales linearly with domain count (≈10 ms per transit domain) due to label-stitching overhead, while SRv6 maintains near-constant latency through single SID substitution, making SRv6 the preferable choice for deployments spanning more than three domains.

ADSO addresses a gap that has remained open since the standardisation of the H-PCE architecture in RFC 8685 [6]: to the best of the authors' knowledge, no prior standard or published proposal provides topology-concealing, sub-50 ms domain state change notifications from Child PCEs to a Parent PCE. The proposed protocol represents a principled and implementable candidate for standardisation as a PCEP extension in the IETF PCE Working Group.

B. Future Work

The present work opens several directions for further investigation:

1) **Kernel MPLS dataplane validation:** The current testbed uses IP-level forwarding as a substitute for kernel MPLS due to namespace limitations in Mininet. Enabling kernel MPLS forwarding via `modprobe mpls_router mpls_iptunnel`, or migrating to Containerlab with FRR containers, will allow measurement of real SR-MPLS dataplane switching latency and produce empirically grounded Table R3 results.
2) **Live PCEP PCInitiate/PCUpdate implementation:** Replacing the simulated PCEP push with actual PCInitiate and PCUpdate messages to a live PCC using the `pcep-lib` Python library or the `pypce` package will eliminate the single remaining simulation dependency and validate the end-to-end PCEP signalling pipeline.
3) **Formal information-theoretic privacy proof:** A rigorous proof that the five ADSO metrics form a sufficient statistic for domain health while satisfying a formal definition of topology indistinguishability (analogous to k-anonymity or differential privacy) is a necessary prerequisite for any security-sensitive deployment.
4) **SRv6 full segment list encoding:** Implementing SRv6 path computation and segment list generation with measured Segment Routing Header overhead will yield directly comparable latency measurements against the SR-MPLS results, validating the theoretical scalability advantage.
5) **Multi-controller scalability study:** Extending the testbed to 5–10 domains with independent OpenDaylight instances per domain will characterise the scalability limits of the Parent PCE's abstract model maintenance and the notification processing throughput under concurrent domain events.
6) **BGP-LS integration with ODL:** Completing the BGP-LS topology feed from FRR ASBR nodes to OpenDaylight using ODL 0.18+ (which corrects the OpenConfig BGP YANG model mismatch present in ODL 0.12.2) will enable the Child PCE to populate ODL's topology datastore and validate the full BGP-LS-to-ADSO information pipeline.
7) **IETF Internet-Draft submission:** Translating the ADSO TLV specification into an IETF Internet-Draft for the PCE Working Group constitutes the immediate next step toward production deployment and operator adoption.

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